



计算机网络实验报告

警示

- 1.实验报告如有雷同，雷同各方当次实验成绩均以 0 分计。
- 2.当次小组成员成绩只计学号、姓名登录在下表中的。
- 3.在规定时间内未上交实验报告的，不得以其他方式补交，当次成绩按 0 分计。
- 4.实验报告文件以 PDF 格式提交。

院系	计算机学院	班 级	计科 1 班	组长	陈安康
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编程实验

【实验内容】

- (1) 完成教材 P233 实例 7-1
- (2) 完成教材 P273 习题 15

学号	学生	自评分
22320021	陈安康	100
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22336070	冯乙山	100
22336064	杜雨彤	100

实验分工见个人心得

【交实验报告】

上传实验报告：助教 截止日期（不迟于）：1 周之内

上传包括两个文件：

(1) 小组实验报告。上传文件名格式：小组号_编程实验.pdf （由组长负责上传）

例如：文件名“10_编程实验.pdf”表示第 10 组的编程实验报告

(2) 小组成员实验体会。每个同学单独交一份只填写了实验体会的实验报告。只需填写自己的学号和姓名。

文件名格式：小组号_学号_姓名_编程实验.pdf （由组员自行上传）

例如：文件名“10_05373092_张三_编程实验.pdf”表示第 10 组的编程实验报告。

注意：不要打包上传！



实验 7-1

步骤 1 (1) 连接后未配置, 无法 ping 通

```
C:\Windows\system32>ping 192.168.3.22

正在 Ping 192.168.3.22 具有 32 字节的数据:
请求超时。
请求超时。
请求超时。
请求超时。

192.168.3.22 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 0, 丢失 = 4 (100% 丢失),

C:\Windows\system32>
```

(2) 路由表

```
管理员: 命令提示符
C:\Windows\system32>route print

=====
接口列表
11...00 0d 0a 4b 18 90 .....Ralink RT61 Turbo Wireless LAN Card
14...00 88 99 00 13 68 .....Realtek PCIe GBE Family Controller
9...18 60 24 8c 17 27 .....Realtek PCIe GBE Family Controller #2
1.....Software Loopback Interface 1
=====

IPv4 路由表
=====
活动路由:
网络目标      网络掩码      网关      接口      跃点数
-----
0.0.0.0        0.0.0.0        172.16.0.1  172.16.4.3  291
0.0.0.0        0.0.0.0        192.168.1.1  192.168.1.11  35
127.0.0.0      255.0.0.0      在链路上    127.0.0.1  331
127.0.0.1      255.255.255.255  在链路上    127.0.0.1  331
127.255.255.255 255.255.255.255 在链路上    127.0.0.1  331
172.16.0.0      255.255.0.0    在链路上    172.16.4.3  291
172.16.4.3      255.255.255.255 在链路上    172.16.4.3  291
172.16.255.255 255.255.255.255 在链路上    172.16.4.3  291
192.168.1.0      255.255.255.0  在链路上    192.168.1.11  291
192.168.1.11     255.255.255.255 在链路上    192.168.1.11  291
192.168.1.255    255.255.255.255 在链路上    192.168.1.11  291
224.0.0.0        240.0.0.0      在链路上    127.0.0.1  331
224.0.0.0        240.0.0.0      在链路上    192.168.1.11  291
224.0.0.0        240.0.0.0      在链路上    172.16.4.3  291
255.255.255.255 255.255.255.255 在链路上    127.0.0.1  331
255.255.255.255 255.255.255.255 在链路上    192.168.1.11  291
255.255.255.255 255.255.255.255 在链路上    172.16.4.3  291
=====

永久路由:
网络地址      网络掩码      网关地址      跃点数      默认
-----
0.0.0.0        0.0.0.0        172.16.0.1      默认
0.0.0.0        0.0.0.0        192.168.1.1      0

20-RSR20-1#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
20-RSR20-1#
```

步骤 3 配置后路由表, 有 S 条目, 通过 ip route 指令设置静态路由

可以看到, 配置后相应接口的网络地址发生变化, 新增一些静态路由条目



IPv4 路由表

活动路由:

网络目标	网络掩码	网关	接口	跃点数
0.0.0.0	0.0.0.0	172.16.0.1		291
0.0.0.0	0.0.0.0	192.168.1.1		25
127.0.0.0	255.0.0.0		在链路上	331
127.0.0.1	255.255.255.255		在链路上	331
127.255.255.255	255.255.255.255		在链路上	331
172.16.0.0	255.255.0.0		在链路上	291
172.16.4.3	255.255.255.255		在链路上	291
172.16.255.255	255.255.255.255		在链路上	291
192.168.1.0	255.255.255.0		在链路上	281
192.168.1.11	255.255.255.255		在链路上	281
192.168.1.255	255.255.255.255		在链路上	281
224.0.0.0	240.0.0.0		在链路上	331
224.0.0.0	240.0.0.0		在链路上	281
224.0.0.0	240.0.0.0		在链路上	291
255.255.255.255	255.255.255.255		在链路上	331
255.255.255.255	255.255.255.255		在链路上	281
255.255.255.255	255.255.255.255		在链路上	291

永久路由:

网络地址	网络掩码	网关地址	跃点数
0.0.0.0	0.0.0.0	172.16.0.1	默认
0.0.0.0	0.0.0.0	192.168.1.1	0

```
Protocol
Serial 2/0                                192.168.2.1/24    no address    up
up
SIC-3G-WCDMA 3/0                          no address      no address    up
down
GigabitEthernet 0/0                       no address      no address    down
down
GigabitEthernet 0/1                       192.168.1.1/24    no address    up
up
VLAN 1                                    no address      no address    up
down
20-RSR20-1(config)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
S    192.168.3.0/24 [1/0] via 192.168.2.2
20-RSR20-1(config)#
```

步骤 6 配置全部完成后，可以 ping 通，执行 traceroute 命令

```
C:\Windows\system32>tracert 192.168.3.22
```

通过最多 30 个跃点跟踪

到 D52_11 [192.168.3.22] 的路由:

```
 1  <1 毫秒  <1 毫秒  <1 毫秒  192.168.1.1
 2  41 ms    41 ms    44 ms    192.168.2.2
 3  48 ms    47 ms    46 ms    D52_11 [192.168.3.22]
```

跟踪完成。

```
C:\Windows\system32>ping 192.168.3.22
```

正在 Ping 192.168.3.22 具有 32 字节的数据:

```
来自 192.168.3.22 的回复: 字节=32 时间=37ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=40ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=40ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=37ms TTL=126
```

192.168.3.22 的 Ping 统计信息:

```
数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
  最短 = 37ms, 最长 = 40ms, 平均 = 38ms
```

```
C:\Windows\system32>
```

通过 wireshark 抓包可以看到 ping 时的连通性



91	20.607746	192.168.1.11	192.168.3.22	ICMP	106 Echo (ping) request id=0x0001, seq=29/7424, ttl=3 (reply in 92)
92	20.656511	192.168.3.22	192.168.1.11	ICMP	106 Echo (ping) reply id=0x0001, seq=29/7424, ttl=126 (request in 91)
93	20.659516	192.168.1.11	192.168.3.22	ICMP	106 Echo (ping) request id=0x0001, seq=30/7680, ttl=3 (reply in 94)
94	20.708519	192.168.3.22	192.168.1.11	ICMP	106 Echo (ping) reply id=0x0001, seq=30/7680, ttl=126 (request in 93)
95	20.711710	192.168.1.11	192.168.3.22	ICMP	106 Echo (ping) request id=0x0001, seq=31/7936, ttl=3 (reply in 96)
96	20.760296	192.168.3.22	192.168.1.11	ICMP	106 Echo (ping) reply id=0x0001, seq=31/7936, ttl=126 (request in 95)
97	20.763759	192.168.1.11	192.168.3.22	NBNS	92 Name query NBSTAT * <00:<00:<00:<00:<00:<00:<00:<00:<00:<00>>
98	20.819398	192.168.3.22	192.168.1.11	NBNS	199 Name query response NBSTAT
99	21.013558	192.168.1.11	192.168.3.22	DNS	85 Standard query 0x0000 AAAA www.bluemagazine.com

- (1) 可能是相应配置问题，如接口未打开，ip 设置错误等。或者接口松动
- (2) show interfaces fastEthernet 端口号

```
show ip route
```

```

20-RSR20-1(config)#show interfaces fastEthernet 1/1
Index(dec):14 (hex):e
FastEthernet 1/1 is DOWN , line protocol is DOWN
Hardware is Broadcom FastEthernet
Interface address is: no ip address
MTU 1514 bytes, BW 100000 Kbit
Encapsulation protocol is Bridge, loopback not set
Keepalive interval is 10 sec , set
Carrier delay is 2 sec
Rxload is 1/255, Txload is 1/255
Queueing strategy: FIFO
Output queue 0/0, 0 drops;
Input queue 0/75, 0 drops
Switchport attributes:
interface's description:"
admin medium-type is Copper, oper medium-type is Copper
lastchange time:0 Day: 0 Hour:31 Minute:28 Second
Priority is 0
admin duplex mode is AUTO, oper duplex is Unknown
admin speed is AUTO, oper speed is Unknown
flow control admin status is ON, flow control oper status is Unknown
Storm Control: Broadcast is ON, Multicast is OFF, Unicast is OFF
Port-type: access
Vlan id: 1
5 minutes input rate 0 bits/sec, 0 packets/sec
5 minutes output rate 0 bits/sec, 0 packets/sec
302 packets input, 136106 bytes, 0 no buffer, 0 dropped
Received 182 broadcasts, 0 runts, 0 giants
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 abort
0 packets output, 0 bytes, 0 underruns , 0 dropped
0 output errors, 0 collisions, 0 interface resets
20-RSR20-1(config)#

```



```
20-RSR20-2#show ip interface
Serial 2/0
  IP interface state is: UP
  IP interface type is: POINTOPOINT
  IP interface MTU is: 1500
  IP address is:
    192.168.2.2/24 (primary)
  IP address negotiate is: OFF
  Forward direct-broadcast is: OFF
  ICMP mask reply is: ON
  Send ICMP redirect is: ON
  Send ICMP unreachable is: ON
  DHCP relay is: OFF
  Fast switch is: ON
  Help address is:
  Proxy ARP is: ON

Serial 3/0
  IP interface state is: DOWN
  IP interface type is: POINTOPOINT
  IP interface MTU is: 1500
  IP address is:
    No address configured
  IP address negotiate is: OFF
  Forward direct-broadcast is: OFF
  ICMP mask reply is: ON
  Send ICMP redirect is: ON
  Send ICMP unreachable is: ON
  DHCP relay is: OFF
  Fast switch is: ON
  Help address is: 0.0.0.0
  Proxy ARP is: ON

GigabitEthernet 0/0
  IP interface state is: DOWN
  IP interface type is: BROADCAST
  IP interface MTU is: 1500
  IP address is:
    No address configured
  IP address negotiate is: OFF
  Forward direct-broadcast is: OFF
  ICMP mask reply is: ON
  Send ICMP redirect is: ON
  Send ICMP unreachable is: ON
  DHCP relay is: OFF
  Fast switch is: ON
  Help address is: 0.0.0.0
  Proxy ARP is: ON

GigabitEthernet 0/1
  IP interface state is: UP
  IP interface type is: BROADCAST
  IP interface MTU is: 1500
```

```
0 output errors, 0 collisions, 0 interface resets
20-RSR20-1#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
S    192.168.3.0/24 [1/0] via 192.168.2.2
20-RSR20-1#
```

(3) 每个路由条目包含哪几项？分别有什么含义？

在网络中，路由条目通常包含以下几项：

目标网络：目标网络是路由条目的目标地址，表示数据包要发送到的网络或主机的地址。

子网掩码：子网掩码用于确定目标网络的范围，帮助路由器识别哪些地址属于同一网络。

下一跳：下一跳是指数据包在转发过程中下一个要发送到的路由器的地址。路由器根据这个地址将数据包发送到正确的方向。

接口：接口是路由器用来发送数据包的物理或虚拟接口。路由器通过这个接口将数据包发送到下一跳或目标网络。

路由类型：路由条目可以是静态路由或动态路由。静态路由是手动配置的，而动



态路由是通过路由协议自动学习的。

度量值：度量值用于评估路由的优劣，通常是一个数值，数值越小表示路由越优先。不同的路由协议可能使用不同的度量标准。

(4) 路由器中如果同时存在去往同一网段的静态路由信息与动态路由信息，路由器会采取哪一个进行转发？

在路由器中，如果同时存在去往同一网段的静态路由信息与动态路由信息，路由器会优先选择静态路由进行转发。这是因为静态路由被认为是管理员手动配置的，通常被视为更可靠的选择。动态路由虽然可以自动更新，但可能会受到网络变化的影响，因此在存在静态路由的情况下，路由器会优先使用静态路由。

习题 15

在如图 7-36 所示的拓扑结构中配置 PC1 到 PC2 之间的静态路由并检查 PC1 与 PC2 的连通性。按顺序完成以下要求：

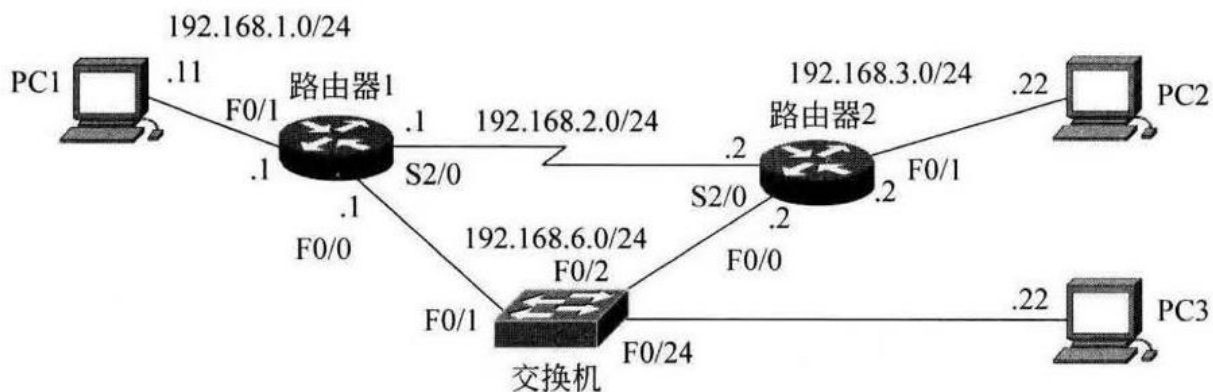


图 7-36 第 15 题拓扑结构

(1) 记录 2 台路由器的路由表。



```
20-RSR20-1#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
S    192.168.3.0/24 [1/0] via 192.168.2.2
```

```
20-RSR20-2(config)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is 192.168.6.1 to network 0.0.0.0
S*   0.0.0.0/0 [1/0] via 192.168.6.1
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.2/32 is local host.
C    192.168.3.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.3.1/32 is local host.
C    192.168.6.0/24 is directly connected, GigabitEthernet 0/0
C    192.168.6.2/32 is local host.
```

(2) 用 PC1 ping PC2, 记录交换机的 MAC 地址表。

Ping 之前的 MAC 表:

20-S5750-1#show mac-address-table			
Vlan	MAC Address	Type	Interface
-----	-----	-----	-----
1	0088.9900.1376	DYNAMIC	GigabitEthernet 0/24

PC1 ping PC2:

```
C:\Windows\system32>ping 192.168.3.22

正在 Ping 192.168.3.22 具有 32 字节的数据:
来自 192.168.3.22 的回复: 字节=32 时间=37ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=40ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=40ms TTL=126
来自 192.168.3.22 的回复: 字节=32 时间=37ms TTL=126

192.168.3.22 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
往返行程的估计时间(以毫秒为单位):
    最短 = 37ms, 最长 = 40ms, 平均 = 38ms
```




Ping 后 MAC 地址:

Vlan	MAC Address	Type	Interface
1	0088.9900.1040	DYNAMIC	GigabitEthernet 0/24
1	1414.4b7b.d9c8	DYNAMIC	GigabitEthernet 0/1
1	1414.4b7b.e3b4	DYNAMIC	GigabitEthernet 0/2

- (3) 清除 MAC 地址表, 启动 Wireshark 捕获, 用 PC1 ping PC2, 查看 PC3 是否可以捕获到 ARP 包、Echo 请求包和 Echo 响应包。记录交换机的 MAC 地址表。

捕获结果:

1	0.000000	192.168.6.22	10.8.8.8	DNS	77 Standard query 0x193f SOA www.microsoft.com
2	0.000332	192.168.6.1	192.168.6.22	ICMP	70 Destination unreachable (Network unreachable)
3	1.033980	192.168.6.22	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
4	2.532364	192.168.6.22	192.168.6.255	UDP	1482 49985 → 1689 Len=1440
5	4.038941	192.168.6.22	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
6	7.042818	192.168.6.22	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
7	11.061148	192.168.6.22	192.168.6.255	UDP	1482 49985 → 1689 Len=1440
8	18.249115	fe80::fad5:6ed7:27e...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
9	18.249200	192.168.6.22	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.0.0.252
10	18.257065	fe80::fad5:6ed7:27e...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
11	18.257138	192.168.6.22	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.0.0.252 for any sources
12	18.257618	192.168.6.22	224.0.0.251	MDNS	72 Standard query 0x0000 ANY D52_10.local, "QM" question
13	18.257924	fe80::fad5:6ed7:27e...	ff02::fb	MDNS	92 Standard query 0x0000 ANY D52_10.local, "QM" question
14	18.258066	192.168.6.22	10.8.8.8	DNS	77 Standard query 0x2ba1 SOA www.microsoft.com
15	18.259121	192.168.6.22	224.0.0.251	MDNS	110 Standard query response 0x0000 AAAA fe80::fad5:6ed7:27ef:ba2f A 192.168.6.22
16	18.259210	fe80::fad5:6ed7:27e...	ff02::fb	MDNS	130 Standard query response 0x0000 AAAA fe80::fad5:6ed7:27ef:ba2f A 192.168.6.22
17	18.259250	fe80::fad5:6ed7:27e...	ff02::1:3	LLMNR	86 Standard query 0xb2ac ANY D52_10
18	18.259326	192.168.6.22	224.0.0.252	LLMNR	66 Standard query 0xb2ac ANY D52_10

无法捕捉到 ARP 包、Echo 请求包和 Echo 响应包

交换机的 MAC 地址表:

Vlan	MAC Address	Type	Interface
1	0088.9900.1376	DYNAMIC	GigabitEthernet 0/24
1	5869.6c27.c09d	DYNAMIC	GigabitEthernet 0/1

- (4) 重新启动 Wireshark 捕获, 用 PC2 ping PC1, 查看是否可以捕获到 ARP 包、Echo 请求包和 Echo 响应包。如果有则对捕获的包截屏。查看并记录 (截屏) PC1 的 ARP 缓冲区。最后, 对结果进行分析。

PC2 ping PC1:

捕获结果:



1	0.000000	192.168.6.22	192.168.6.255	UDP	1482 49985 → 1689 Len=1440
2	2.300756	fe80::fad5:6ed7:27e...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
3	2.300776	192.168.6.22	224.0.0.22	IGMPv3	54 Membership Report / Leave group 224.0.0.252
4	2.316473	fe80::fad5:6ed7:27e...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
5	2.316562	192.168.6.22	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.0.0.252 for any sources
6	2.317051	192.168.6.22	224.0.0.251	MDNS	72 Standard query 0x0000 ANY D52_10.local, "QM" question
7	2.317053	192.168.6.22	10.8.8.8	DNS	77 Standard query 0xb881 SOA www.microsoft.com
8	2.317170	fe80::fad5:6ed7:27e...	ff02::fb	MDNS	92 Standard query 0x0000 ANY D52_10.local, "QM" question
9	2.317255	192.168.6.22	224.0.0.251	MDNS	110 Standard query response 0x0000 AAAA fe80::fad5:6ed7:27ef:ba2f A 192.168.6.22
10	2.318336	fe80::fad5:6ed7:27e...	ff02::fb	MDNS	130 Standard query response 0x0000 AAAA fe80::fad5:6ed7:27ef:ba2f A 192.168.6.22
11	2.318398	fe80::fad5:6ed7:27e...	ff02::1:3	LLMNR	86 Standard query 0xe5e1 ANY D52_10
12	2.318462	192.168.6.22	224.0.0.252	LLMNR	66 Standard query 0xe5e1 ANY D52_10
13	2.329541	192.168.6.1	192.168.6.22	ICMP	70 Destination unreachable (Network unreachable)
14	2.370268	192.168.6.22	239.255.255.250	SSDP	179 M-SEARCH * HTTP/1.1
15	2.429089	192.168.6.22	224.0.0.22	IGMPv3	54 Membership Report / Join group 224.0.0.252 for any sources
16	2.429127	fe80::fad5:6ed7:27e...	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
17	3.317813	192.168.6.22	10.8.8.8	DNS	77 Standard query 0xb881 SOA www.microsoft.com

并没有捕获到 ARP 包、Echo 请求包和 Echo 响应包。

查看 ARP 缓冲区：

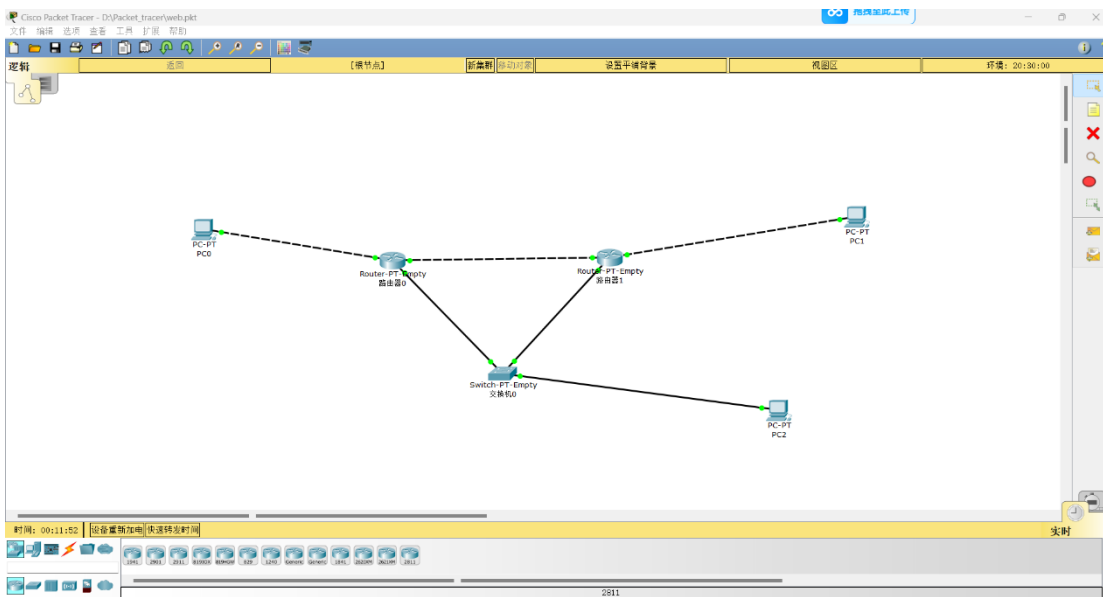
```
:\Windows\system32>arp -a

接口: 172.16.4.3 --- 0x9
Internet 地址      物理地址      类型
172.16.0.1          a4-bb-6d-e2-5a-34 动态
172.16.0.231        e8-fb-1c-d3-ee-ed 动态
172.16.11.1         18-60-24-88-a9-79 动态
172.16.12.2         18-60-24-8c-17-10 动态
172.16.20.1         18-60-24-8c-17-16 动态
172.16.255.255      ff-ff-ff-ff-ff-ff 静态
224.0.0.22          01-00-5e-00-00-16 静态
224.0.0.251         01-00-5e-00-00-fb 静态
224.0.0.252         01-00-5e-00-00-fc 静态
234.2.2.4           01-00-5e-02-02-04 静态
239.255.255.250     01-00-5e-7f-ff-fa 静态

接口: 192.168.1.11 --- 0xe
Internet 地址      物理地址      类型
192.168.1.1         58-69-6c-27-c0-9e 动态
192.168.1.255       ff-ff-ff-ff-ff-ff 静态
224.0.0.2           01-00-5e-00-00-02 静态
224.0.0.22          01-00-5e-00-00-16 静态
224.0.0.251         01-00-5e-00-00-fb 静态
224.0.0.252         01-00-5e-00-00-fc 静态
239.255.255.250     01-00-5e-7f-ff-fa 静态
```

因为此时还没有配置端口镜像，且静态路由配置为两个路由器直接相连，所以数据包直接从路由器 2 发送到路由器 1，不经过交换机，在 PC3 上捕获不到数据包。

(5) 在 Packet Tracer 上配置网络拓扑图如下：



其中两个路由器的静态路由配置如下：

路由器1

物理配置CLI属性

全局
设置
运算规则设置
路由
静态
RIP
接口
FastEthernet0/0
FastEthernet1/0
FastEthernet2/0

静态路由

网络192.168.1.0
掩码255.255.255.0
下一跳192.168.2.1

添加

网络地址

192.168.1.0/24 via 192.168.2.1

删除

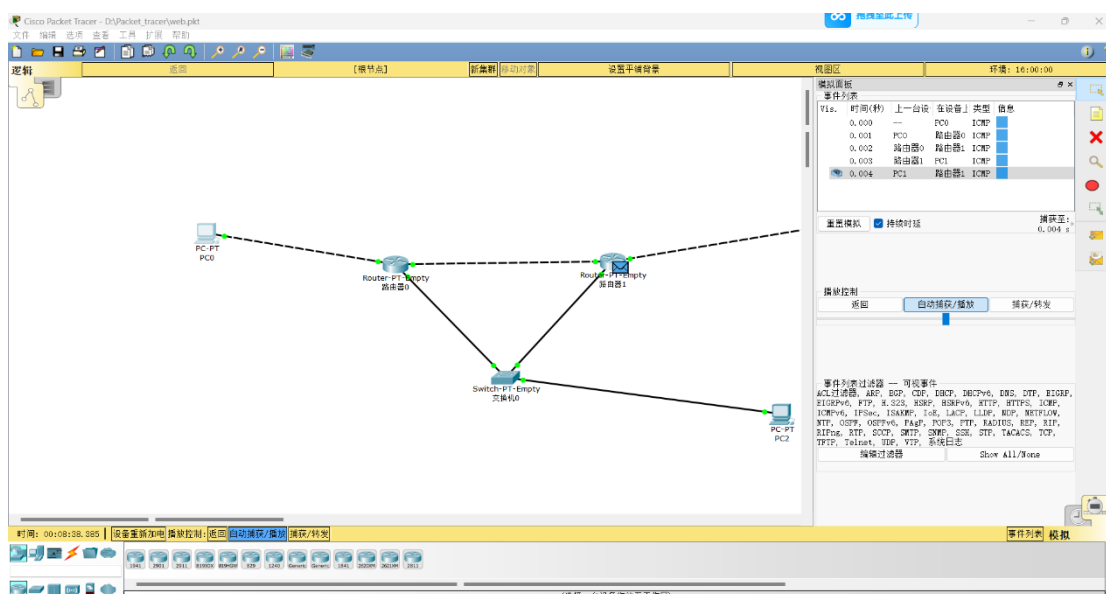
等效的IOS命令

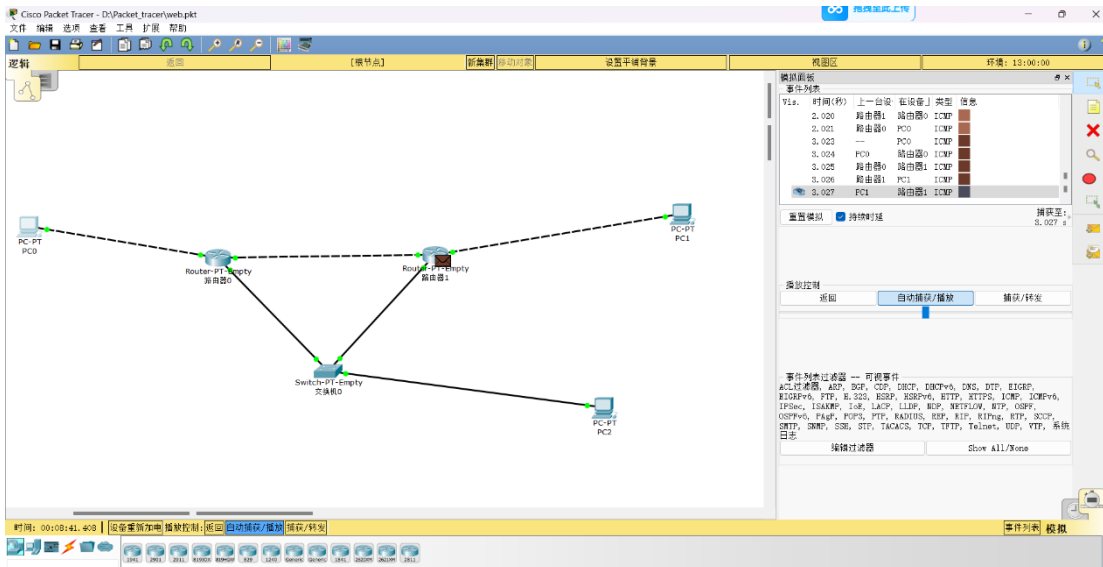
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.2.1
Router(config)#
```

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启动 Flash 动画功能，过程截图如下：





PC 的命令行界面结果如下：

```
PC0
物理 配置 桌面 属性 自定义界面

命令提示符

Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.22

Pinging 192.168.3.22 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.3.22:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>ping 192.168.3.22

Pinging 192.168.3.22 with 32 bytes of data:

Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.3.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>
```

经过 Flash 动画模拟我们发现，ICMP 包只通过路由器之间的线路传输，而不经交换机（见下图）。



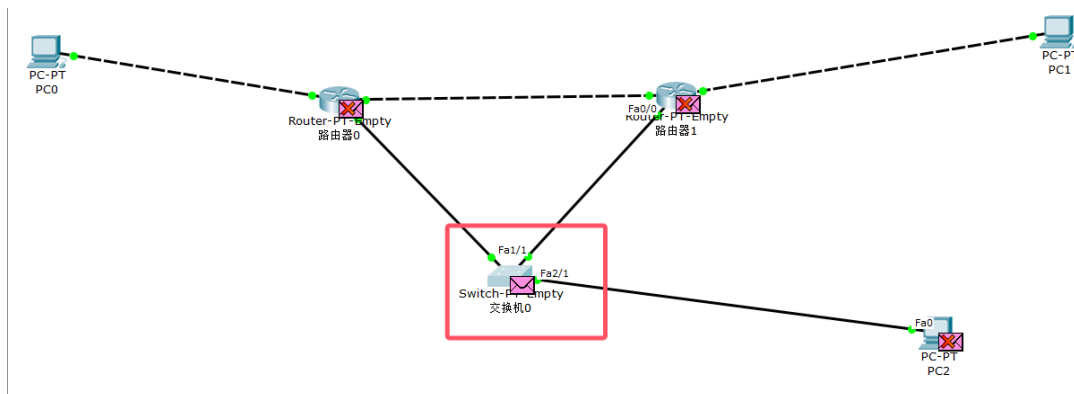
计算机网络实验报告

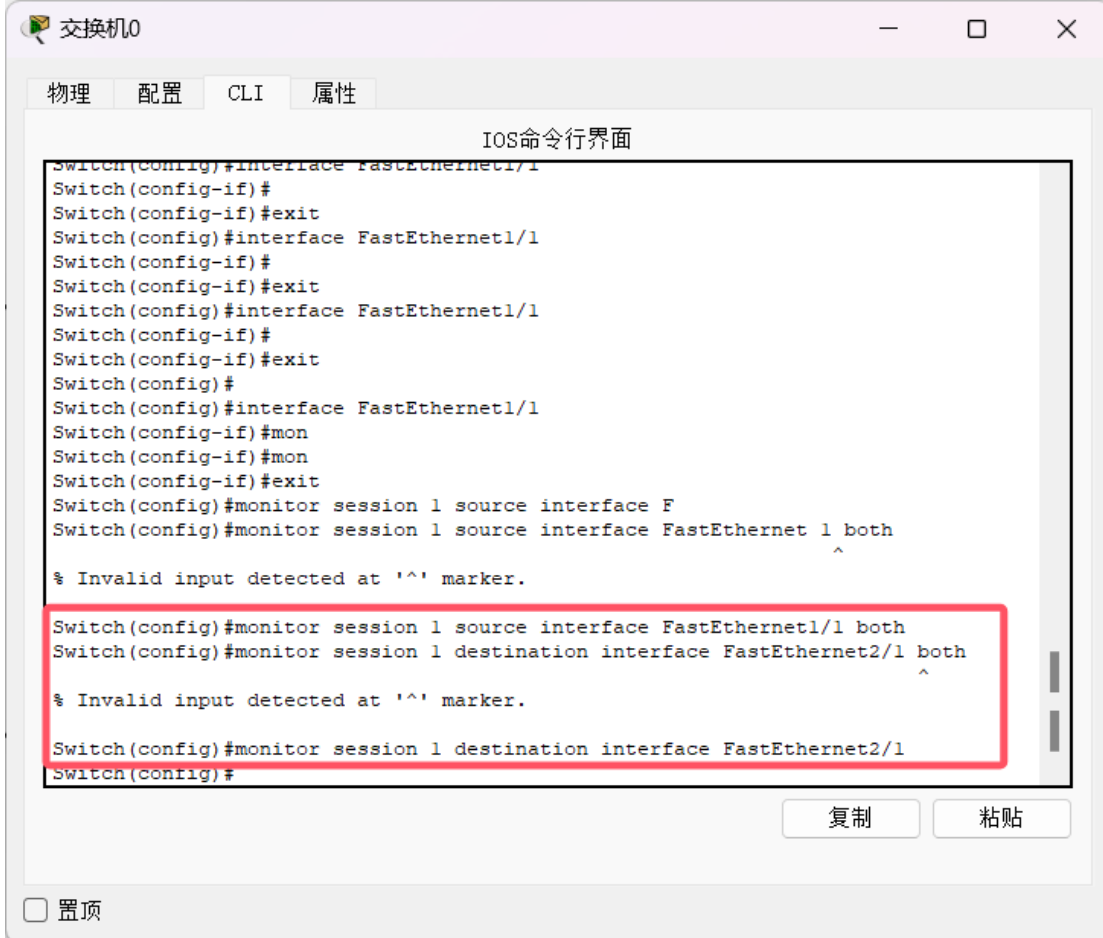
这是因为我们在两个路由器之间配置了静态路由，在路由选择时，路由器会首先查找静态路由表，如果找到匹配的静态路由，则会使用该静态路由进行数据包的转发；如果没有找到匹配的静态路由，路由器才会考虑使用动态路由协议学习到的路由信息。

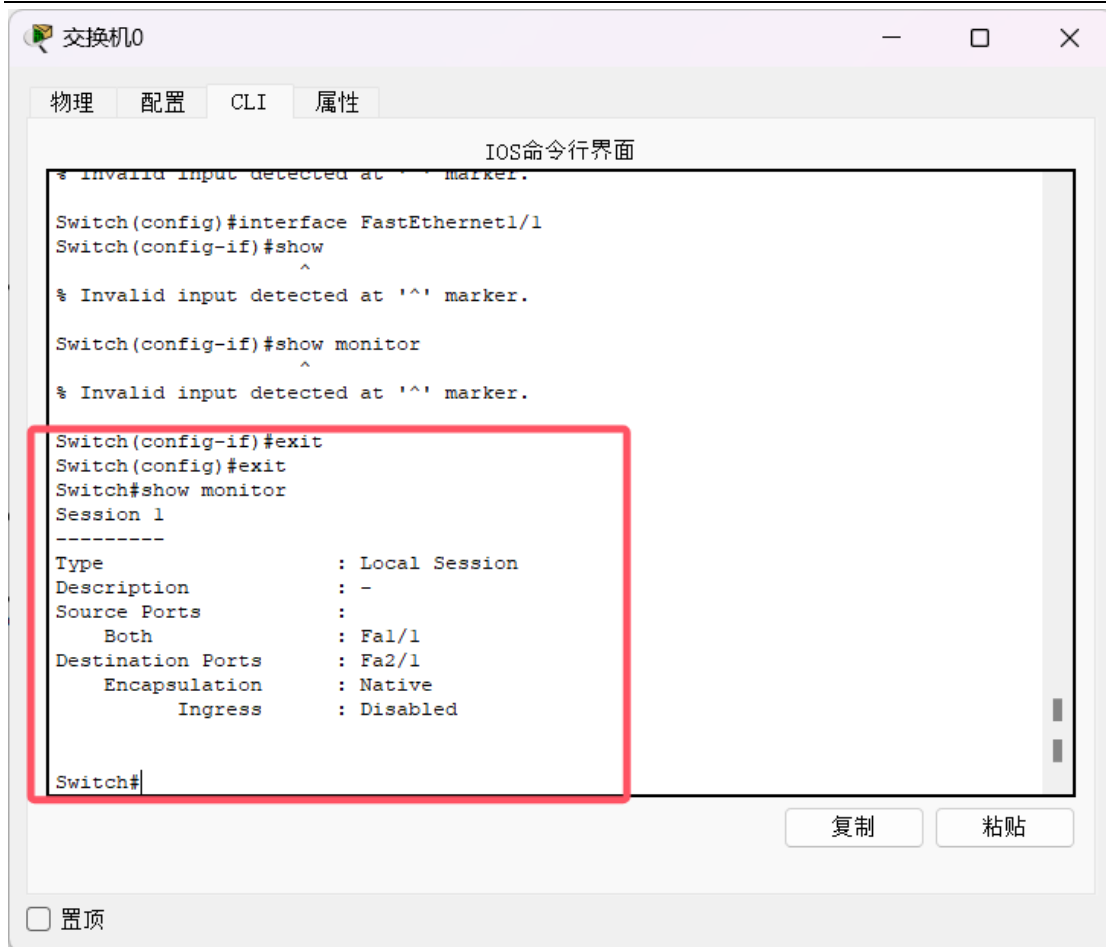
Vis.	时间(秒)	上一台设备	在设备上	类型	信息
	0.000	--	PC0	ICMP	
	0.001	PC0	路由器0	ICMP	
	0.002	路由器0	路由器1	ICMP	
	0.003	路由器1	PC1	ICMP	
	0.004	PC1	路由器1	ICMP	
	0.005	路由器1	路由器0	ICMP	
	0.006	路由器0	PC0	ICMP	

(6) 将交换机端口 0/24 镜像到 0/24，后，令 PC1 ping PC2 PC3 启动抓包，结果并没有捕获到相应的包，查看交换机的 MAC 表并没有记录 MAC 地址，这是因为配置镜像端口会自动清除 MAC 地址，而由于静态路由的存在，ICMP 包并不会经过交换机的端口，自然 PC3 不会抓到任何有用的数据包，MAC 地址表也为空。

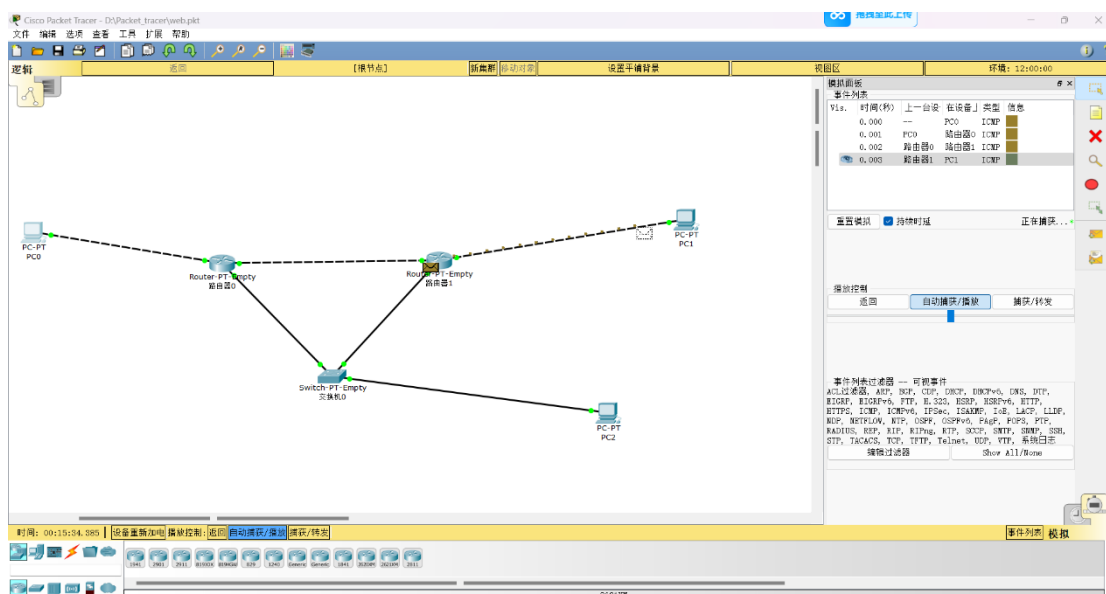
(7) 在 Packet Tracer 中找到两个端口的名称，配置镜像端口：

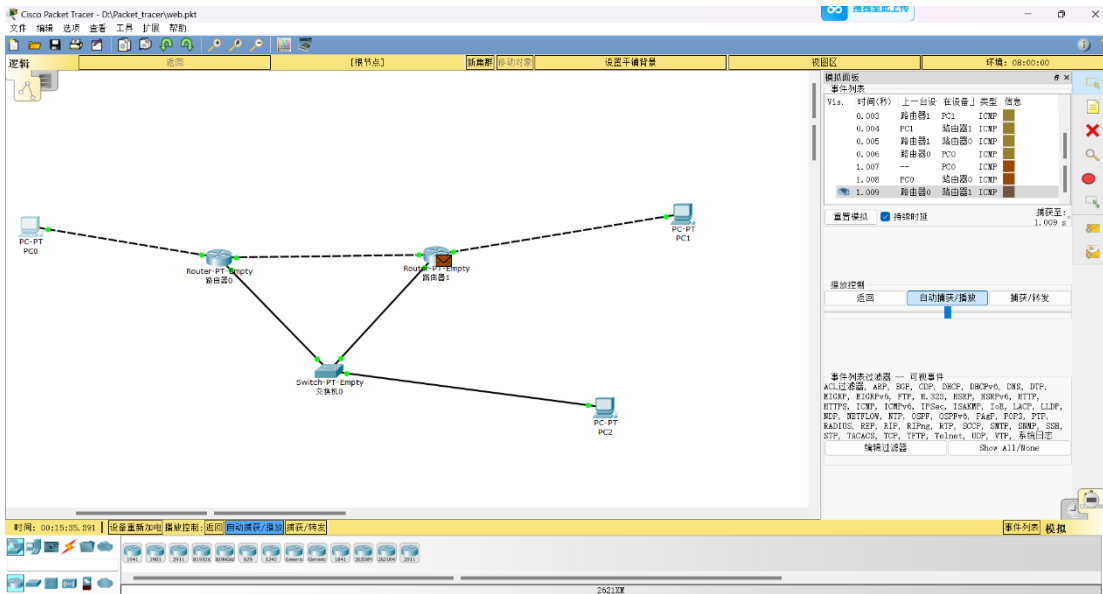






启动 Flash 动画模式，过程如下：





PC0

物理 配置 桌面 属性 自定义界面

命令提示符

```
Ping statistics for 192.168.3.22:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>ping 192.168.3.22

Pinging 192.168.3.22 with 32 bytes of data:

Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.3.22:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>ping 192.168.3.22

Pinging 192.168.3.22 with 32 bytes of data:

Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126
Reply from 192.168.3.22: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.3.22:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 6ms, Average = 6ms

C:\>
```

☐ 置顶



Vis.	时间(秒)	上一台设备	在设备上	类型	信息
	1.007	--	PC0	ICMP	
	1.008	PC0	路由器0	ICMP	
	1.009	路由器0	路由器1	ICMP	
	1.010	路由器1	PC1	ICMP	
	1.011	PC1	路由器1	ICMP	
	1.012	路由器1	路由器0	ICMP	
	1.013	路由器0	PC0	ICMP	

由上图可知，数据包依旧在配置过静态路由的链路上通过，而不经交换机，与我们预期一致。

(8) 由于静态路由的存在，数据不经过交换机，镜像端口不能抓到包。

(9) 在路由器 1 上删除静态路由，配置默认路由如下：

```
20-RSR20-1(config)#no ip route 192.168.3.0 255.255.255.0 192.168.2.2
20-RSR20-1(config)#show ip route
% Unknown command.

20-RSR20-1(config)#show ip route
% Unknown command.

20-RSR20-1(config)#exit
20-RSR20-1#*Nov 13 18:26:04: %SYS-5-CONFIG_I: Configured from console by console
20-RSR20-1#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is no set
C    192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C    192.168.1.1/32 is local host.
C    192.168.2.0/24 is directly connected, Serial 2/0
C    192.168.2.1/32 is local host.
C    192.168.6.0/24 is directly connected, GigabitEthernet 0/0
C    192.168.6.1/32 is local host.
20-RSR20-1#
```



```
20-RSR20-1#ip route 0.0.0.0 0.0.0.0 192.168.6.2
% Unknown command.

20-RSR20-1#confi
20-RSR20-1#configure ter
20-RSR20-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
20-RSR20-1(config)#
20-RSR20-1#Nov 13 18:28:15: %SYS-5-CONFIG_I: Configured from console by console

20-RSR20-1#
20-RSR20-1#
20-RSR20-1#con
20-RSR20-1#configure ter
20-RSR20-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
20-RSR20-1(config)#ip route 0.0.0.0 0.0.0.0 192.168.6.2
20-RSR20-1(config)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is 192.168.6.2 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 192.168.6.2
C 192.168.1.0/24 is directly connected, GigabitEthernet 0/1
C 192.168.1.1/32 is local host.
C 192.168.2.0/24 is directly connected, Serial 2/0
C 192.168.2.1/32 is local host.
C 192.168.6.0/24 is directly connected, GigabitEthernet 0/0
C 192.168.6.1/32 is local host.
20-RSR20-1(config)#
```

21	24.726115	192.168.1.11	192.168.3.22	IOP	74 Echo (ping) request	id=0x0001, seq=25/6400, ttl=127 (no response found)
23	25.739604	192.168.1.11	192.168.3.22	IOP	74 Echo (ping) request	id=0x0001, seq=26/6656, ttl=127 (no response found)
24	26.745000	192.168.1.11	192.168.3.22	IOP	74 Echo (ping) request	id=0x0001, seq=27/6912, ttl=127 (no response found)
25	27.750832	192.168.1.11	192.168.3.22	IOP	74 Echo (ping) request	id=0x0001, seq=28/7168, ttl=127 (no response found)

此时只能捕获请求包，不能捕获应答包。这是因为因为路由器 1 删除了静态路由后，数据包全部经过交换机到达 PC2，因此配置了镜像端口的 PC3 可以捕获到请求包，而由于 PC2 的

静态路由此时还没有被删除，因此应答包还是经过路由器之间的链路进行传输，因此镜像端口捕获不到应答包。

然后删除路由器 2 的静态路由，配置默认路由如下：



```
20-RSR20-2#
20-RSR20-2#
20-RSR20-2#con
20-RSR20-2#configure ter
20-RSR20-2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
20-RSR20-2(config)#no ip route 192.168.1.0 255.255.255.0 192.168.2.1
20-RSR20-2(config)#ip route 0.0.0.0 0.0.0.0 192.168.6.1
20-RSR20-2(config)#show ip route

Codes: C - connected, S - static, R - RIP, B - BGP
        O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default

Gateway of last resort is 192.168.6.1 to network 0.0.0.0
S* 0.0.0.0/0 [1/0] via 192.168.6.1
C 192.168.2.0/24 is directly connected, Serial 2/0
C 192.168.2.2/32 is local host.
C 192.168.3.0/24 is directly connected, GigabitEthernet 0/1
C 192.168.3.1/32 is local host.
C 192.168.6.0/24 is directly connected, GigabitEthernet 0/0
C 192.168.6.2/32 is local host.
```

此时 PC1 ping PC2, PC3 进行抓包, 可以顺利抓到 Echo 包:

132 0.268223	192.168.1.11	192.168.3.22	ICMP	74 Echo (ping) request	id=0x0001, seq=33/8448, ttl=127 (reply in 133)
133 0.268224	192.168.3.22	192.168.1.11	ICMP	74 Echo (ping) reply	id=0x0001, seq=33/8448, ttl=127 (request in 132)
135 0.634723	192.168.6.2	192.168.1.11	ICMP	70 Redirect	(Redirect for host)
163 0.635441	192.168.6.2	192.168.1.11	ICMP	70 Redirect	(Redirect for host)
165 0.635441	192.168.6.2	192.168.1.11	ICMP	70 Redirect	(Redirect for host)
167 0.635441	192.168.6.2	192.168.1.11	ICMP	70 Redirect	(Redirect for host)
168 0.635441	192.168.6.2	192.168.1.11	ICMP	70 Redirect	(Redirect for host)
169 0.635441	192.168.6.2	192.168.1.11	ICMP	70 Redirect	(Redirect for host)
258 1.256421	192.168.3.22	192.168.6.2	ICMP	74 Echo (ping) request	id=0x0001, seq=34/8784, ttl=127 (reply in 269)
269 1.257432	192.168.3.22	192.168.1.11	ICMP	74 Echo (ping) reply	id=0x0001, seq=34/8784, ttl=127 (request in 268)
270 2.272937	192.168.1.11	192.168.3.22	ICMP	74 Echo (ping) request	id=0x0001, seq=35/8960, ttl=127 (reply in 271)
271 2.273947	192.168.3.22	192.168.1.11	ICMP	74 Echo (ping) reply	id=0x0001, seq=35/8960, ttl=127 (request in 270)

由于静态路由全部被删除, 因此应答包和请求包全部走默认路由, 经过交换机, 镜像端口因此全部可以捕获到。

(10) 当 ping 一个拓扑结构之外的 IP 地址 (192.168.250.250), 抓包如下:

No.	Time	Source	Destination	Protocol	Length	Info
127	3.017205	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
128	3.017329	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
129	3.017329	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
130	3.017331	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
131	3.017331	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
132	3.017331	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
133	3.017673	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
134	3.017693	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
135	3.017693	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
136	6.544292	00:00:00:00:00:13:68	00:00:00:00:00:13:68	ARP	42	Who has 192.168.1.1? Tell 192.168.1.11
137	6.544483	192.168.1.11	192.168.250.250	ICMP	74	Echo (ping) request id=0x0001, seq=42/10752, ttl=128 (no response found!)
138	6.544708	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
139	6.544708	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
140	6.544708	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
141	6.544720	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
142	6.544720	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
143	6.544720	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
144	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
145	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
146	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
147	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
148	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
149	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
150	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
151	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
152	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
153	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
154	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
155	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
156	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
157	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
158	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
159	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
160	6.545237	192.168.6.2	192.168.1.11	ICMP	70	Redirect (Redirect for host)
161	6.545237	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
162	6.545823	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
163	6.545823	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
164	6.545823	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
165	6.545823	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)
166	6.545823	192.168.1.1	192.168.1.11	ICMP	70	Redirect (Redirect for host)

可以看到有大量数据包由路由器 1, 路由器 2 发送过来, 这是因为未知数据包会发送往



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默认路由，但路由器 2 不知道这个包要发往哪里，它也会发往默认路由，这样就会形成很多数据包在之间来回传送。